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1 Set & class theory

Notation:

Starting here and for the rest of the document, we will use usual symbol of equality "=" for quasi-equality (before it was "$\simeq$") and "$\equiv$" for equality (before it was "=").

Definition 1.1. Identity.

$$I = \{(x, x) | \top\}$$

Theorem 1.1. Identity is a relation.

$$(I \text{ relation})$$

$$(I \subseteq V \times V)$$

Definition 1.2. "F on X".

$$\text{"F on X"} = F \restriction X$$

Definition 1.3.

$$(y \in f[A]) \longleftrightarrow (\exists x \in A : \langle x, y \rangle \in f)$$

Definition 1.4.

$$(y \in f\langle A \rangle) \longleftrightarrow (\forall x \in A : \langle x, y \rangle \in f)$$

Theorem 1.2.

$$X \subseteq \mathcal{P}Y \longleftrightarrow \bigcup X \subseteq Y$$

Theorem 1.3.

$$(g \text{ functional}) \wedge (M \text{ countable}) \longrightarrow (g[M] \text{ countable})$$

Theorem 1.4.

$$(f^{\leftarrow} \text{ univocal})$$

Theorem 1.5.

$$(f \text{ injective}) \longrightarrow \left(f\left[\bigcup M\right] = \bigcup f^{\rightarrow}[M]\right)$$

Theorem 1.6.

$$(f \text{ univocal}) \longrightarrow f^{-1}\left[\bigcup M\right] = \bigcup f^{\leftarrow}[M]$$